

Hybrid Solar Decision Layer for Bangladesh

SolarThik

Part 1: SME Market Opportunity & Financial Case

Poultry and cold storage as the lead SME verticals, residential/commercial retained

Prepared July 2026 — Working strategy document, not a fundraising deck or financial offer
Combines Sprint 1 (SME market analysis) and Sprint 2 (cost-benefit financial models).
Sprint 3 (software architecture) to follow as Part 2.

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Executive Summary

Two SME verticals were tested against real, sourced load-profile and market data: **poultry (broiler sheds)** and **cold storage (potato/produce)**. A third candidate the founders raised — generic “waste processing” — did not hold up as a distinct, sizeable Bangladeshi SME category once tested, and is addressed honestly in Part A3 rather than forced into the plan.

The single most important finding of this phase: **for both SME verticals, a simple “solar savings beat your EMI” pitch does not clear on the numbers.** The real economics come from a different place in each case — avoided catastrophic loss for poultry (heat-stress mortality), and avoided spoilage plus time-of-use tariff arbitrage for cold storage. This is a genuinely useful finding, not a setback: it tells us precisely what the software layer has to do in Part 2 (Sprint 3) — model and price risk, not just meter kilowatt-hours.

Headline numbers

Poultry: a documented 15% flock-loss event covers **~14 months** of system financing in one avoided incident.

Cold storage: a documented spoilage event, scaled to a 3,000-tonne facility, covers **~37–50 months** of financing. Residential/commercial (built in Sprint 0) remains the only segment where bill-savings alone already clear the EMI — at ~8 of 12 months covered by default, improving with a bigger down payment or longer tenure.

Recommendation carried into Part 2: price and pitch poultry and cold storage as a **resilience/insurance product with a financing wrapper**, not a bill-savings product with a resilience feature. The monitoring and risk-scoring layer is the product for these two segments — not an add-on to it.

Part A — SME Market Analysis

A1. Poultry — the strongest fit of the three candidates

Bangladesh has an estimated **100,000 commercial poultry farms** (roughly 60,000 broiler, 23,000 layer, 13,000 Sonali/colour-bird; older industry sources put total farmer count nearer 150,000 — treat as a range). Total sector investment is around **\$4.1–4.2 billion**, employing 6–8.5 million people directly and indirectly. Critically for go-to-market: **53.3% of farms sit in Dhaka division, with 24% concentrated specifically in Gazipur district** — a natural, dense first-territory. 17–19% of production already runs through contract-farming integrator arrangements, meaning a handful of integrators sit on top of dozens or hundreds of individual sheds — a far more efficient channel than farm-by-farm sales.

This is a reliability sale, not a cost sale. In April 2024, a Narayanganj farmer reported **over 500 chickens dead from heatstroke in two sheds** during sustained load-shedding; the president of the Bangladesh Poultry Association said he was receiving reports of heat-stroke deaths **from farms nationwide**, with egg production down 15–30% and broilers finishing roughly 500g lighter during these events. Most sheds are tin-roofed with high stocking density and minimal airflow — described in the literature as becoming “heat chambers” the moment ventilation stops. The sector already has a documented financing gap: industry sources note many small and medium farms have abandoned poultry altogether due to a lack of financing and high lending rates.

A2. Cold storage — the best-validated segment researched

Around **400 cold storage facilities** operate nationally with roughly 6 million tonnes of combined capacity, mostly for potatoes. Bangladesh produces over 10 million tonnes of potatoes annually against under 4 million tonnes of storage capacity — a structural shortage that supports continued sector growth independent of anything we do.

The trade association has already made the pitch for us

The Bangladesh Cold Storage Association has formally and publicly demanded electricity tariffs be cut from Tk13.62/unit (peak) and Tk9.62/unit (off-peak) to Tk5, and loan interest rates cut from 17% to 7%. Those are the exact two variables — power cost and financing cost — a solar-plus-EMI product is built to attack.

The failure mode is already happening and already quantified: a 2023 Daily Star report described 400 facilities nationwide struggling through repeated power cuts, with one 10,000-tonne facility alone estimated to have lost **Tk1.5–2 crore** from reduced capacity. One important caution, because rigor cuts both ways: this segment is already financially stressed — roughly **300 of the 400 facilities nationally cannot repay their bank loans on time**. That is a strong argument for staying a packager here rather than a lender — that credit risk belongs with lenders who already underwrite this sector, not on our own book. The model is already proven at small scale: one 3,000-tonne facility profile describes solar backup cutting power bills by roughly 25%, and a Bangladesh Agricultural University prototype (HortiCool) already runs a small cold-storage unit on solar with LiFePO4 battery backup and grid as last resort — the exact architecture we would be proposing, already built and tested domestically.

A3. Waste processing — an honest dead end, and the real alternative

Flag

This sub-section exists specifically because the brief was “do not hallucinate or give vague work.” “Waste processing” did not survive contact with real data as a distinct SME category, and it is reported here honestly rather than papered over with an invented market size.

The well-documented “waste-to-energy” story in Bangladesh is municipal, not SME: the Aminbazar plant in Dhaka is a **\$300 million, 42.5MW facility** built by a Chinese state EPC (CMEC) under a 25-year power purchase agreement with BPDB, consuming 3,000 tonnes/day of municipal solid waste. That is utility-scale infrastructure competing in the arena the Part 1 memo already flagged as the wrong one for a startup to enter — not an SME opportunity, and we should not force it into one.

The one legitimate SME-adjacent thread is **effluent treatment plants (ETPs)** — mandatory wastewater treatment attached to garment, textile, and tannery factories, running pumps and aerators continuously, with passing evidence of some existing solar use in factory operations data. This is a real, compliance-driven, power-intensive SME category structurally similar to the net-metering mandate already identified — but it needs its own dedicated research pass (sizing, load profile, tariff class) before it is trusted the way poultry and cold storage now are. **Recommendation: re-rank the target list as poultry, cold storage, and ETP-attached garment/textile facilities** — with ETP explicitly flagged as needing another research pass, not yet a proven third pillar.

A4. Cross-cutting go-to-market playbook

- **Geographic concentration is a gift.** Both poultry (Gazipur/Dhaka division) and much of the cold storage industry (documented concentration in Bogura, Kurigram, Gaibandha, Munshiganj) cluster regionally — build a dense first territory rather than a thin national footprint.
- **Trade associations and integrators are the channel, not individual SMEs.** The Bangladesh Cold Storage Association and Bangladesh Poultry Association are organized, vocal, and already making the exact argument this product answers. Poultry integrators controlling dozens of contract-farmed sheds are a single sales conversation that unlocks many customers.
- **Sell resilience/insurance first, cost second.** Both segments have a real, documented, recurring catastrophic-failure mode. That is the opening line, not the bill-savings calculator.
- **Stay a packager, not a lender, especially in cold storage,** where ~75% of existing facilities are already loan-distressed. Let existing sector lenders hold that risk; supply the feasibility and monitoring data that makes them comfortable extending it.

Part B — Cost-Benefit Financial Models

Built on the same tariff/EMI engine as the Sprint 0 residential/commercial calculator, with load profiles grounded in sourced engineering data rather than invented consumption figures. Every assumption is labeled as sourced or modeled — see callouts and the methodology section.

B1. Poultry shed model — 2,000-bird broiler house

Load basis: a Nigerian commercial-poultry sizing guide (500–2,000 birds — closely matching Bangladesh's documented 500–2,500-bird typical farm size) puts peak simultaneous load at 2,000–3,000W during brooding (heat lamps + fans + lights together), dropping to 800–1,500W continuous outside brooding. A broiler cycle runs ~35–42 days grow-out plus ~10–14 days cleanout, giving roughly 7 flocks/year.

Input	Value	Basis
Brooding load (first ~14 days/cycle)	2.5 kW	Sourced range midpoint (2.0–3.0kW)
Normal grow-out load	1.15 kW	Sourced range midpoint (0.8–1.5kW)
Heat-event load (stress-test)	2.1 kW	Modeled from CFM/kg ventilation ratios — flagged as derived, not direct
Blended monthly consumption	~985 kWh/mo	Weighted by cycle-phase share of the year
Proposed system	3.0 kW solar + 10 kWh battery	Battery sized to ride ~4.8h of a heat-event outage
System cost	Tk 535,000	Tk95/W solar (Tk285,000) + Tk25,000/kWh battery (Tk250,000, illustrated)

The honest result: bill savings alone do not clear the EMI

Season	Solar savings/mo	EMI/mo (15% down, 6yr, 11%)	Net position
Dry season	Tk 5,827	Tk 8,656	– Tk 2,828
Average month	Tk 4,819	Tk 8,656	– Tk 3,837
Monsoon (worst)	Tk 3,698	Tk 8,656	– Tk 4,958

This is a small system serving a small load — the battery needed for real resilience is nearly half the total cost, and it does not itself generate bill savings. Sold purely as “cut your power bill,” this product fails on the numbers. It should not be sold that way.

The real pitch: avoided catastrophic loss

Mortality event	Birds lost	Value lost (Tk180/kg, 2.2kg avg)	Months of EMI covered
10%	200	Tk 79,200	9.1 months
15% (documented-scale event)	300	Tk 118,800	13.7 months
25%	500	Tk 198,000	22.9 months
50%	1,000	Tk 396,000	45.7 months
Full flock (worst case)	2,000	Tk 792,000	91 months

Reference case

A Narayanganj farmer reported 500+ birds dead from heatstroke across two sheds during load-shedding in April 2024 (The Business Standard); the national Poultry Association president reported similar losses being called in from farms nationwide that same season. This is a recurring seasonal risk (April–June), not a rare tail event — broiler farm-gate prices (Tk120–230/kg over 2026, highly volatile) mean even a partial loss is a large single-event cost relative to the financing.

B2. Cold storage model — 3,000-tonne potato facility

Load basis: industry benchmarks put efficient cold-storage energy intensity at 560–700 kWh/tonne/year; Bangladeshi facilities are largely older ammonia plants without VFD optimization, so 800 kWh/tonne/year is used as a “typical, not-yet-efficient” planning figure — above the efficient benchmark, flagged as an estimate. Tariff figures (Tk13.62 peak / Tk9.62 off-peak) are directly sourced from the Bangladesh Cold Storage Association's own public statements.

Metric	Value	Basis
Annual consumption	2,400,000 kWh	3,000t × 800 kWh/t/yr (modeled)
Average continuous load	274 kW	Compressors run 18–24h/day per BD sources
Blended tariff	Tk 10.45/unit	5h peak (Tk13.62) / 19h off-peak (Tk9.62) — split is modeled, rates
Estimated current annual bill	Tk 25,088,000	Consumption × blended tariff

Solar offset is structurally limited here — and that's an honest finding

System size	Generation (avg mo)	% of annual load offset
100 kW (conservative)	10,458 kWh	5.2%
250 kW	26,144 kWh	13.1%
400 kW (roof-maximized)	41,830 kWh	20.9%

Even a large, roof-maximized system only dents a fraction of total consumption, because compressors run 18–24 hours a day — a load with none of the day/night shape that makes solar work well for poultry or residential customers. A 100kW system costs roughly **Tk8,500,000**; at 20% down, 7-year tenure, 12% APR, the EMI (Tk120,039/mo) exceeds even the conservative-case monthly savings (Tk100,602), a net of **–Tk19,436/month**. Pure kWh-offset is the wrong pitch for this segment.

Two real value levers instead

- **Time-of-use arbitrage (a software problem, not a solar problem).** Shifting ~1,370 kWh/day out of the Tk13.62 peak window into battery/solar-sourced supply, at the Tk4.00/unit rate spread, is worth roughly **Tk166,575/month** — independent of how much solar is actually installed. This is exactly the kind of dispatch-optimization job the monitoring/control software layer should own.
- **Avoided spoilage loss.** A documented 2023 case saw a 10,000-tonne facility lose Tk1.5–2 crore from a single sustained power-cut event. Scaled illustratively to our 3,000-tonne facility: **Tk4,500,000–6,000,000** — covering 37–50 months of the solar system's own financing in one avoided incident.

Caution

~300 of Bangladesh's 400 cold storage facilities cannot currently repay their bank loans on time. This segment's underlying credit risk is real and already elevated — strong reason to stay a packager here, supplying the feasibility case to existing agri-lenders rather than carrying this credit risk on our own book.

B3. Residential/commercial model — retained baseline

The generic residential/commercial calculator built in Sprint 0 is not being dropped — SMEs are the lead focus going forward, but residential remains part of the product and is the one segment where a straightforward bill-savings pitch already works on its own terms.

Metric	Value
Profile	Commercial connection, 800 units/month, 5.5kWp system
System cost / EMI	Tk 495,000 total → Tk 8,009/month (15% down, 6yr, 11% APR)
Savings — dry / average / monsoon	Tk 10,684 / Tk 8,835 / Tk 6,780
Net — dry / average / monsoon	+Tk 2,675 / +Tk 826 / –Tk 1,229
Months of the year savings ≥ EMI	8 of 12 (improves with larger down payment or longer tenure)
Simple payback	4.64 years

B4. Cross-segment summary

Segment	Bill-savings-only verdict	Real value driver	Underwriting logic
Residential/commercial	Works (8/12 months by default)	Direct cost offset	Standard EMI-vs-bill
Poultry	Fails (net negative all seasons)	Avoided flock-mortality loss	Insurance-style: price against avoided-loss
Cold storage	Fails, and structurally limited	Peak-shaving arbitrage + avoided spoilage	Price arbitrage + dispatch-optimization value, not bill-savings

This is the central takeaway to carry into the software architecture phase: **the two SME verticals both require the platform to model and price risk, not just meter and bill energy.** That has direct implications for what the software must do — covered in Part 2.

Bridge to Part 2 — Software Architecture

This phase surfaced specific, concrete requirements the software layer will need to satisfy, which Part 2 (Sprint 3) will architect against directly:

- A **risk-scoring engine** for poultry (flock-loss probability by season/region) and cold storage (spoilage-event probability), not just a kWh feasibility calculator.
- A **battery dispatch controller** that optimizes against time-of-use tariffs for cold storage (peak-shaving), and against outage-survival duration for poultry — two different optimization objectives, same underlying control problem.
- A **monitoring layer that feeds an underwriting data trail** — generation, discharge events, and near-miss outage rides-through — to build the credibility with lenders/insurers that both segments' trade associations have said is currently missing.
- Integration with **integrator/association-level reporting**, not just single-site dashboards, given the channel strategy runs through integrators and trade bodies rather than individual SMEs.

Methodology & Sources

Market sizing and sector data: One Health Poultry Hub, Bangladesh Poultry Industries Central Council and Bangladesh Poultry Association public statements, The Business Standard, The Daily Star, The Financial Express, eFeedLink. Cold storage: Bangladesh Cold Storage Association public statements, The Daily Star, Bangladesh Agricultural University (HortiCool prototype), "Project Profile Bangladesh" facility examples. Waste-to-energy: reporting on the Aminbazar WtE plant (Dhaka). Engineering/load-profile benchmarks: Auburn University and University of Georgia poultry ventilation extension guides, Volcano Electric EC-fan sizing guide, a Nigeria-market commercial poultry inverter-sizing guide (used for small-farm load bands, given closer scale match to Bangladesh's 500–2,500-bird typical farm than large US tunnel-house examples), and Rinac (cold-storage energy-efficiency benchmarks).

Where a figure is explicitly modeled rather than directly sourced (e.g., the heat-event load estimate, the peak/off-peak hour split, the battery cost per kWh, the linear scaling of the documented spoilage-loss case), it is labeled as such at point of use. Broiler and potato prices are both volatile — ranges are shown rather than single points where the underlying data supports a range. Nothing in this document is a loan offer, financial advice, or an engineering specification — every load and cost figure needs verification against a real site audit, installer quote, and lender term sheet before customer-facing use.